

University of South Carolina
Department of Civil and Environmental Engineering
ECIV 542: Traffic Engineering

Homework #1

Due: 10:59 AM, Sep 9, 2026

Guideline for HW:

- Clearly show your work for the following HW problems
- Use the “HW Submission Template” to solve the problems
- Print/scan your work (in pdf) format and upload it to Blackboard
- Late homework submissions will **NOT** be accepted.

1. A driver is traveling at 55 mph on a road that is level. At one point, he sees an overturned truck at 350 feet ahead. If the driver is able to decelerate at a rate of 10 ft/s^2 , (a) at what speed will his vehicle hit the truck? (b) Use Excel to plot the result for reaction times ranging from 0.5 to 5.0 seconds in increments of 0.5 seconds. Comment on the results.

2. A car hits a tree at an estimated speed of 25 mph on a 3% grade. If skid marks of 150 feet are observed on dry pavement (side friction factor = 0.35) followed by 250 feet on a grassy shoulder (side friction factor = 0.2), estimate the initial speed of the vehicle just before the driver slams on the brakes.

3. How long should the yellow interval be for vehicles approaching a traffic signal on a 2% downgrade at a speed of 40 mph? Use a standard reaction time of 1.0 second and 10 ft/s^2 deceleration rate. Determine if there is a dilemma zone if the all-red interval is set at 1.5 seconds and intersection width is 48 feet. If yes, how long? Assume a standard vehicle length of 20 feet.

4. A 3.50 horizontal curve is to be designed on a highway with a design speed of 60 mi/hr. Spiral transition curves are to be used. Determine the length of the spiral and the appropriate stations for the T.S., S.C., C.S., and S.T. The angle of deflection for the original tangents is 40° , and the PI is at station $151 + 26$.

5. What length of superelevation runoff should be used to achieve a superelevation rate of 10%? The design speed is 70 mph, and a three-lane cross section (12-ft lanes) is under consideration. Superelevation will be achieved by rotating all three lanes around the inside edge of the pavement.

6. A vertical curve of 1,000 feet is designed to connect a grade of +4% to a grade of -5%. The PVI is at station $15 + 55$ and at an elevation of 500 feet. Find the following:

- a) The station of the PVC and PVT
- b) The elevation of the PVT and PVI
- c) The elevation of points along the vertical curve at 100-ft intervals
- d) The location and elevation of the high point on the curve